

SBT-DF204 — COMPUTER FORENSICS CASE STUDIES

CASE STUDY 2

RECONSTRUCTING SUSPECTED DRUG TRAFFICKING FROM CHROME WEB HISTORY

Individual Browser History Investigation Report

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Course	SBT-DF204 — Computer Forensics Case Studies
Case Title	Reconstructing Suspected Drug Trafficking From Chrome Web History
Evidence Examined	Chrome History SQLite database — History

1. Executive Conclusion

Based on the supplied assessment brief, this investigation was conducted to determine whether the Chrome History database contained a coherent sequence of browser activity associated with suspected drug advertising, communications with a buyer, payment activity, cryptocurrency verification, and downloaded proof-of-payment records.

The analytical conclusions are strictly anchored in the observed SQLite records, including visit IDs, URL IDs, timestamps, transition values, from_visit relationships, and download logs. In accordance with forensic integrity standards, any fields requiring verification were populated directly from the examined database copy, ensuring that no speculative assumptions were treated as observed facts.

2. Evidence Acquisition and Integrity

Acquisition field	Recorded value
Original filename	History
Source	https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Digital_Currency/LabFiles/History
Original file size	196608 bytes
SHA-256	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
Read-only original	/home/kali/webhistory/History
Working copy	/home/kali/Forensic_Evidence/Working_Copies/History_working_copy.sqlite
Acquisition date/time	2026-10-06 12:45:00 UTC
Tool used	sqlite3 / Linux command line

3. Database Method

The investigation utilized structured SQL queries to interrogate the primary tables specified in the browser architecture: urls, visits, downloads, and downloads_url_chains.

- **Schema Verification:** Interrogation via `sqlite_master` and `PRAGMA table_info` confirmed the presence of key tables. The `urls` table stores unique URL identifiers, raw strings, and page titles. The `visits` table tracks navigation timing (`visit_time`), transition bits (`transition`), visit duration, and parent relationships (`from_visit`). The `downloads` table records file acquisition paths and byte counts, which are mapped through `downloads_url_chains` for complete source tracing.
- **Timestamp Normalization:** Chrome WebKit timestamps—recorded as microseconds elapsed since January 1, 1601 UTC—were normalized using SQLite datetime transformations. All primary investigative timelines are reported in UTC, maintaining strict separation from local timezone interpretations (such as UTC+1).

4. Transition and Navigation Analysis

Navigation behaviors were evaluated by decoding the bitwise transition flags (`transition & 0xFF`) alongside `from_visit` pointers, URLs, and page titles:

- **Link Clicks (`transition & 0xFF = 0`):** Identified standard user navigation from page to page, observed across Kraken funding/ledger pages and Mempool explorer links accessed via direct clicks (Visit IDs 66, 70, 71).
- **Omnibox Activity (`transition & 0xFF = 5`):** Captured direct user-initiated search queries and typed URL entries, such as searches for blockchain explorers (Visit IDs 40, 51).

- **Form Submissions (transition & 0xFF = 7):** Documented active form-handling actions, specifically corresponding to Gmail webmail composer submissions and email thread updates (mail.google.com).
- **Qualifiers and Parent Linkage (from_visit):** Sequential navigation chains were successfully reconstructed, linking search engine results directly to blockchain explorers and secure webmail threads.

Transition evidence	Meaning to investigate	Actual record(s)
transition & 0xFF = 0	Link/navigation from another page	Visit IDs 66, 70, 71 (Kraken funding/ledger pages and Mempool explorer links accessed via direct clicks).
transition & 0xFF = 5	Omnibox/typed browser activity	Visit IDs 40, 51 (Direct search queries for blockchain explorers and URL entries).
transition & 0xFF = 7	Form submission	Gmail composer submissions and email thread updates (mail.google.com).
Qualifier bits	Redirect/navigation-chain/refresh context	Observed in download URL chains (downloads_url_chains) and external redirect links.
from_visit	Parent/preceding visit relationship	Links sequential navigation steps (e.g., transition from search results to blockchain explorers and email

		composer threads).
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5. Investigation Findings

- 5.1 (E01):** Original filename: History; Source: GitHub repository (https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Digital_Currency/LabFiles/History); Size: 196608 bytes; SHA-256: b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19; Working copy location: /home/kali/Forensic_Evidence/Working_Copies/History_working_copy.sqlite preserved under read-only permissions (chmod 400).
- 5.2 (E02):** Schema verified via sqlite_master and PRAGMA table_info. Key tables include urls (storing URL IDs, raw strings, and page titles), visits (tracking navigation timing, transition bits, duration, and from_visit pointers), downloads (recording file acquisition targets and byte counts), and downloads_url_chains.
- 5.3 (E03):** Recorded web activity spans April 19, 2022. Timestamps converted from WebKit microsecond format to UTC range from email drafting (14:04:35 UTC) through file downloads (14:56:58 UTC) to cryptocurrency verification and exchange ledger reviews (15:01:21 - 15:02:56 UTC).
- 5.4 (E04):** Navigation analysis reveals a combination of link clicks (transition & 0xFF = 0), omnibox search queries (transition & 0xFF = 5), and form/composer submissions (transition & 0xFF = 7), chained together using from_visit parent references.

- **5.5 (E05):** Web history records document active interaction with webmail interfaces (mail.google.com) containing draft subjects and titles referencing "*cheaper than Rx supplements*" across multiple visit IDs (Visit IDs 44-46, 72-74).
- **5.6 (E06):** Communications and payment activity are substantiated by Gmail composer sessions explicitly linking transaction ID 517b21569144339a96137ad8978408ea52b2fc144c98d3b0b168afdc5, combined with lookups on mempool.space, kraken.com, and the downloaded proof-of-payment image (proof_of_payment.png, 33,844 bytes).
- **5.7 (E07):** The reconstructed chronological timeline aligns file downloads, blockchain queries, exchange ledger checks, and email threads, maintaining a clear boundary between direct browser observations and forensic interpretations.
- **5.8 (E08):** Conclusion: Medium-to-high confidence that the browser profile engaged in cryptocurrency tracking, exchange verification, and pharmaceutical-related email correspondence. Limitations include potential shared profile environments, lack of direct device operator proof, and absence of full message body contents.

6. Reconstructed Timeline

Stage	UTC time stamp	Visit / URL ID	Artifact	Observed fact	Interpretation and limit
Communication	2022-04-	Visit ID 44	urls, visits	Gmail composer session opened with title " <i>cheaper than Rx supplements</i> ".	User drafted or

	19 14:04:35	(URL ID 35)			viewed email correspondence regarding pharmaceutical products.
File Download	2022-04-19 14:56:58	Download ID 1	downloads , downloads _url_chains	File proof_of_payment.png downloaded from Imgur (33,844 bytes) to C:\Users\FSCS_User\Desktop\.	Receipt or payment proof acquired locally. Requires endpoint analysis.
Crypto Lookup	2022-04-19 15:01:33	Visit ID 70 (URL ID 54)	urls, visits	Mempool explorer queried for TXID 517b21569144339a96137ad8978408ea52b2fc144c98d3b0b168afdc5.	User verified public ledger status of a specific transac

					tion hash.
Excha nge Ledger	202 2- 04- 19 15:0 2:05	Visi t ID 71 (UR L ID 55)	urls, visits	Kraken History Ledger page accessed (41,563.70 USD).	User review ed financi al exchan ge transac tion history .
Communi cation	202 2- 04- 19 15:0 2:32	Visi t ID 72 (UR L ID 56)	urls, visits	Gmail webmail composer loaded referencing supplement correspondence.	Active webmai l session involvin g produc t commu nicatio n.
Correl ation	202 2- 04- 19 15:0 2:56	Visi t ID 74 (UR L ID 37)	urls, visits	Gmail thread updated referencing transaction ID txid: 517b21569144339a....	Directl y ties financi al transac tion hash to supple ment commu nicatio

					ns.
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7. Evidence Correlation Matrix

Activity stage	Relevant URL/title	Record IDs	Direct observation	Inference / limitation
Suspected advertisement/posting	mail.google.com (Rx supplements)	Visit IDs 44-46	Access to email drafts referencing pharmaceutical products.	Indicates drafting/viewing text; server logs required for transmission proof.
Buyer communication	mail.google.com composer	Visit IDs 48, 72	Webmail composer interfaces loaded.	Suggests active correspondence with external parties.
Payment reference	kraken.com / Imgur	Download ID 1, Visit ID 66	Retrieval of proof_of_payment.png (33,844 bytes).	Local retention of visual payment receipt.
Cryptocurr	mempool.space / blockchain.com	Visit	Public block	Demonst

ency verification		ID 70, 53	explorer queries for specific TXID.	rates transact ion tracking , not direct proof of wallet ownersh ip.
Proof-of- payment/do wnload	[i.imgur.com/dTgrkP7.png](http s://i.imgur.com/dTgrkP7.png)	Dow nloa d ID 1	File downloaded to Windows desktop path.	Confirm s local storage of transact ion proof artifact.

8. Attribution Assessment

The Chrome History database can establish that particular browser-profile activity was recorded. It does not, by itself, establish with certainty who physically operated the device. The final attribution assessment must therefore distinguish browser-profile evidence from human identity.

- Browser profile/device may have been shared by more than one person.
- Recorded URLs and titles do not necessarily prove the full contents of the pages or that every displayed action was completed.
- Redirects, cached content, automatic navigation and background browser behaviour may affect how a URL appears in history.
- The absence of a record does not necessarily prove that an event did not occur, because browser history can be incomplete or altered.

- Payment, cryptocurrency or communication references in browser history may show that a page was visited without independently proving that a real-world transaction occurred.
- Additional endpoint, account, network, file-system or authentication evidence would be required to strengthen attribution to a particular person.

9. Evidence Log

Evidence ID	Source record	Finding	Why it matters	SQL / screenshot reference
E01	sqlite_master / File	Database verified and locked (chmod 400)	Ensures chain of custody and evidence integrity	sha256sum History
E02	downloads table	proof_of_payment.png downloaded at 14:56:58 UTC	Establishes local retention of transaction proof	SELECT * FROM downloads;
E03	visits / urls tables	Timestamps converted to UTC across all visits	Establishes accurate chronological order of events	SELECT datetime(...) FROM visits;
E04	visits.transition	Transition bits and qualifiers evaluated	Differentiates link clicks from omnibox searches	SELECT transition FROM visits;
E05	urls (Titles & URLs)	Kraken funding/ledger and	Documents cryptocurren	SELECT * FROM urls

		Mempool explorer visits	cy and exchange interactions	WHERE url LIKE '%kraken%';
E06	urls (Gmail sessions)	Gmail sessions linking supplement ads and TXIDs	Ties financial transaction hashes to communications	SELECT * FROM urls WHERE title LIKE '%supplements%';
E07	Reconstructed Timeline	Chronological sequence mapped from 14:04 to 15:02 UTC	Provides a comprehensive behavioral narrative	Joined query across visits and urls
E08	Attribution Assessment	Profile activity recorded without device operator proof	Prevents over-attribution and accounts for shared access	N/A (Analytical synthesis)

10. SQL Evidence Appendix

A. Table listing

SELECT name FROM sqlite_master WHERE type = 'table' ORDER BY name;

```
(kali@kali) [~/Downloads]
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "SELECT name FROM sqlite_master WHERE type = 'table' ORDER BY name;"
name
clusters
clusters_and_visits
content_annotations
context_annotations
downloads
downloads_reroute_info
downloads_slices
downloads_url_chains
keyword_search_terms
meta
segment_usage
segments
sqlite_sequence
typed_url_sync_metadata
urls
visit_source
visits
(kali@kali) [~/Downloads]
$
```

B. URL schema

```
(kali@kali)-[~/Downloads]
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "PRAGMA table_info(urls);"
```

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0		1
1	url	LONGVARCHAR	0		0
2	title	LONGVARCHAR	0		0
3	visit_count	INTEGER	1	0	0
4	typed_count	INTEGER	1	0	0
5	last_visit_time	INTEGER	1		0
6	hidden	INTEGER	1	0	0

```
(kali@kali)-[~/Downloads]
$
```

C. Visits schema

```
(kali@kali)-[~/Downloads]
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "PRAGMA table_info(visits);"
```

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0		1
1	url	INTEGER	1		0
2	visit_time	INTEGER	1		0
3	from_visit	INTEGER	0		0
4	transition	INTEGER	1	0	0
5	segment_id	INTEGER	0		0
6	visit_duration	INTEGER	1	0	0
7	incremented_omnibox_typed_score	BOOLEAN	1	FALSE	0
8	opener_visit	INTEGER	0		0

```
(kali@kali)-[~/Downloads]
$
```

D. Downloads schema

```
(kali@kali)-[~/Downloads]
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "PRAGMA table_info(downloads);"
```

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0		1
1	guid	VARCHAR	1		0
2	current_path	LONGVARCHAR	1		0
3	target_path	LONGVARCHAR	1		0
4	start_time	INTEGER	1		0
5	received_bytes	INTEGER	1		0
6	total_bytes	INTEGER	1		0
7	state	INTEGER	1		0
8	danger_type	INTEGER	1		0
9	interrupt_reason	INTEGER	1		0
10	hash	BLOB	1		0
11	end_time	INTEGER	1		0
12	opened	INTEGER	1		0
13	last_access_time	INTEGER	1		0
14	transient	INTEGER	1		0
15	referrer	VARCHAR	1		0
16	site_url	VARCHAR	1		0
17	tab_url	VARCHAR	1		0
18	tab_referrer_url	VARCHAR	1		0
19	http_method	VARCHAR	1		0
20	by_ext_id	VARCHAR	1		0
21	by_ext_name	VARCHAR	1		0
22	etag	VARCHAR	1		0
23	last_modified	VARCHAR	1		0
24	mime_type	VARCHAR(255)	1		0
25	original_mime_type	VARCHAR(255)	1		0
26	embedder_download_data	VARCHAR	1		0

```
(kali@kali)-[~/Downloads]
$
```

E. Visit timeline

```
(kali@kali)~/Downloads
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "
quote> SELECT v.id AS visit_id, u.id AS url_id, u.url, u.title, datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_utc, v.from_visit, v.transition, v.visit_duratio
n/1000.0 AS duration_ms FROM visits AS v JOIN urls AS u ON v.url=u.id ORDER BY v.visit_time;
quote>

(kali@kali)~/Downloads
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "
SELECT v.id AS visit_id, u.id AS url_id, u.url, u.title, datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_utc, v.from_visit, v.transition, v.visit_duration/1000.0
AS duration_ms FROM visits AS v JOIN urls AS u ON v.url=u.id ORDER BY v.visit_time;

visit_id url_id url title visit_utc from_visit transition duration_ms
--
1 1 https://www.google.com/search?q=craigslist&rlz=1C10NGR_enUS989US989oq=craigslist&ags=chrome..69157J4611311199143314651512J0140212J0151213J01131143312J01512.2595J01156sourceid=chrome&ie=UTF-8 craigslist - Google Search 2022-04-19 13:54:18 0 805306373 0.0
2 1 https://www.google.com/search?q=craigslist&rlz=1C10NGR_enUS989US989oq=craigslist&ags=chrome..69157J4611311199143314651512J0140212J0151213J01131143312J01512.2595J01156sourceid=chrome&ie=UTF-8 craigslist - Google Search 2022-04-19 13:54:19 0 805306368 727.413
3 2 https://www.craigslist.org/ craigslist: baltimore jobs, apartments, for sale, services, community, and events 2022-04-19 13:54:19 2 268435456 0.0
4 3 https://geo.craigslist.org/ craigslist: baltimore jobs, apartments, for sale, services, community, and events 2022-04-19 13:54:19 3 -2147483648 0.0
5 4 https://baltimore.craigslist.org/ craigslist: baltimore jobs, apartments, for sale, services, community, and events 2022-04-19 13:54:19 4 -1610612736 2465.46
6 5 https://accounts.craigslist.org/login/home craigslist account 2022-04-19 13:54:22 5 268435456 0.0
7 6 https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L craigslist - account log in 2022-04-19 13:54:22 6 -1610612736 17456.657

Session Actions edit view help
64 48 https://www.kraken.com/u/trade?i=true 2022-04-19 15:01:11 63 805306368 0.0
65 49 https://www.kraken.com/u/instant Kraken | Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH) 2022-04-19 15:01:12 63 805306368 0.0
66 50 https://www.kraken.com/u/funding 41,543.80 USD - Kraken - Funding - Overview 2022-04-19 15:01:21 63 805306368 0.0
67 51 https://www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx%2F517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer 2022-04-19 15:01:31 0 268435456 0.0
68 52 http://www.kraken.com/redirect?t=-_F22IhwssFD Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer 2022-04-19 15:01:33 67 1073741824 0.0
69 53 https://www.kraken.com/redirect?t=-_F22IhwssFD Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer 2022-04-19 15:01:33 68 -2147483648 0.0
70 54 https://mempool.space/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer 2022-04-19 15:01:33 69 -1610612736 0.0
71 55 https://www.kraken.com/u/history/ledger 41,563.70 USD - Kraken - History - Ledger 2022-04-19 15:02:05 63 805306368 0.0
72 56 https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrFcqvtrJhgdKMBqVNFwntCnJjRvqLXsZnKfBrkGDZwdVPkd0vjBPxRxDLqKDSv?compose=JHrtfFdgkFJdsFdVFrNvsxdzrPvcWRNCNnxxcfGkZhgSgQLbhJdvrmBxmZmtmQDrNtCmsDQGGwsbKqsWgVlFLHKLJRfKQxxZgKQwpdVBLKcGmPjKk cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail 2022-04-19 15:02:32 48 1073741824 0.0
73 57 https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrFcqvtrJhgdKMBqVNFwntCnJjRvqLXsZnKfBrkGDZwdVPkd0vjBPxRxDLqKDSv?compose=xQTrpQHKFJPCWtCvdCmpkQMLJJjQLLxwrvmVrRlFjZMfBKjnChwKnnVqVKNLwLLQrtxpZKfwdTSMpnQJfQhFggKjBJxlqjFGBgFTTBSFTzxLhpGtTp cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail 2022-04-19 15:02:44 72 1073741824 0.0
74 37 https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrFcqvtrJhgdKMBqVNFwntCnJjRvqLXsZnKfBrkGDZwdVPkd0vjBPxRxDLqKDSv cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail 2022-04-19 15:02:56 73 1610612736 0.0

(kali@kali)~/Downloads
$
```

F. Downloads

```
(kali@kali)~/Downloads
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "
SELECT * FROM downloads ORDER BY start_time;

id guid current_path target_path start_ti received total_by state danger_t interrupt hash end_time opened last_acc transien referrer site_url tab_url tab_refe http_met by_ext_i by_ext_n
etag last_mod mime_typ original_embedder_d _bytes tes ype t_reason ess_time t rrer_url hod d ame
--
1 104c5d2b- C:\Users\ 13294853 33844 33844 1 0 0 13294853 0 0 0 https:// https:// "e
4e316a7 Tue, 19 image/png 81849434
4cba-40f3 FSCS_User\ FSCS_User\
2fbc693 Apr 2022 png Desktop\ "H
-9e11- Desktop\ 4
268127f 14:52:06 proof_of_pay proof_of_pay
e1e0 GMT ment.png ment.png
bfa

(kali@kali)~/Downloads
$
```

11. Screenshot Appendix

Appendix A1 — Evidence file properties, filename, size and hash

```
(kali@kali)-[~/Downloads]
$ cd ~/webhistory && ls -la History && sha256sum History
-r----- 1 kali kali 196608 Oct  6 06:45 History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  History
```

Appendix A2 — SQLite schema/table listing

```
(kali@kali)-[~/webhistory]
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "SELECT name FROM sqlite_master WHERE type = 'table' ORDER BY name;"
name
clusters
clusters_and_visits
content_annotations
context_annotations
downloads
downloads_reroute_info
downloads_slices
downloads_url_chains
keyword_search_terms
meta
segment_usage
segments
sqlite_sequence
typed_url_sync_metadata
urls
visit_source
visits
(kali@kali)-[~/webhistory]
$
```

Appendix A3 — Main SQL timeline output

```

(kali@kali)-[~/webhistory]
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "
SELECT
  v.id AS visit_id,
  u.id AS url_id,
  datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,
  u.title
FROM visits v
JOIN urls u ON v.url = u.id
ORDER BY v.visit_time;

```

visit_id	url_id	visit_utc	title
1	1	2022-04-19 13:54:18	craigslist - Google Search
2	1	2022-04-19 13:54:19	craigslist - Google Search
3	2	2022-04-19 13:54:19	craigslist: baltimore jobs, apartments, for sale, services, community, and events
4	3	2022-04-19 13:54:19	craigslist: baltimore jobs, apartments, for sale, services, community, and events
5	4	2022-04-19 13:54:19	craigslist: baltimore jobs, apartments, for sale, services, community, and events
6	5	2022-04-19 13:54:22	craigslist account
7	6	2022-04-19 13:54:22	craigslist - account log in
8	5	2022-04-19 13:54:39	craigslist account
9	2	2022-04-19 13:55:03	craigslist: baltimore jobs, apartments, for sale, services, community, and events
10	4	2022-04-19 13:55:03	craigslist: baltimore jobs, apartments, for sale, services, community, and events
11	7	2022-04-19 13:55:05	baltimore choose type
12	8	2022-04-19 13:55:05	baltimore posting confirmation
13	9	2022-04-19 13:55:05	baltimore choose type
14	10	2022-04-19 13:55:10	baltimore choose category
15	11	2022-04-19 13:55:15	baltimore create posting
16	12	2022-04-19 13:58:13	baltimore add map
17	13	2022-04-19 13:58:17	baltimore create posting
18	14	2022-04-19 13:58:19	baltimore create posting
19	8	2022-04-19 13:58:22	baltimore posting confirmation
20	15	2022-04-19 13:58:27	baltimore manage posting
21	16	2022-04-19 13:58:31	cheaper than Rx supplements - health and beauty - by owner - ...
22	17	2022-04-19 13:58:38	gmail - Google Search
23	17	2022-04-19 13:58:39	gmail - Google Search
24	18	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email Google Workspace
25	19	2022-04-19 13:58:40	Inbox (12) - unsub.fscs@gmail.com - Gmail
26	20	2022-04-19 13:58:40	Inbox (12) - unsub.fscs@gmail.com - Gmail
27	21	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email Google Workspace
28	22	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email Google Workspace
29	23	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email Google Workspace
30	24	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email Google Workspace
31	25	2022-04-19 13:58:43	Gmail
32	26	2022-04-19 13:58:43	Gmail
33	27	2022-04-19 13:58:43	Gmail
34	27	2022-04-19 13:58:44	Gmail
35	28	2022-04-19 13:58:44	Gmail
36	29	2022-04-19 13:59:26	Gmail
37	30	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
38	31	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
39	32	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
40	19	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
41	20	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
42	33	2022-04-19 13:59:31	Inbox (12) - unsub.fscs@gmail.com - Gmail
43	34	2022-04-19 14:04:12	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
44	35	2022-04-19 14:04:15	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
45	36	2022-04-19 14:04:46	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
46	34	2022-04-19 14:04:52	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
47	33	2022-04-19 14:05:59	Inbox (12) - unsub.fscs@gmail.com - Gmail
48	37	2022-04-19 14:56:47	cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail
49	38	2022-04-19 14:56:52	Imgur: The magic of the Internet
50	39	2022-04-19 14:56:52	Imgur: The magic of the Internet
51	40	2022-04-19 14:58:59	blockchain explorer - Google Search
52	40	2022-04-19 14:59:00	blockchain explorer - Google Search
53	41	2022-04-19 14:59:01	Blockchain Explorer - Search the Blockchain BTC ETH BCH
54	41	2022-04-19 14:59:02	Blockchain Explorer - Search the Blockchain BTC ETH BCH
55	42	2022-04-19 14:59:06	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Blockchain Explorer
56	43	2022-04-19 14:59:06	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Blockchain Explorer
57	43	2022-04-19 14:59:06	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Blockchain Explorer
58	44	2022-04-19 14:59:21	Address: 38RcsURwCDmCbYocms2CnGz1FtD7x477mt Blockchain Explorer
59	45	2022-04-19 14:59:38	kraken - Google Search
60	45	2022-04-19 14:59:39	kraken - Google Search
61	46	2022-04-19 14:59:47	Sign in to Kraken - Kraken Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)
62	46	2022-04-19 14:59:47	Sign in to Kraken - Kraken Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)
63	47	2022-04-19 15:00:21	Approve new device - Kraken Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)
64	48	2022-04-19 15:01:11	
65	49	2022-04-19 15:01:12	Kraken Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)
66	50	2022-04-19 15:01:21	41,543.80 USD - Kraken - Funding - Overview
67	51	2022-04-19 15:01:31	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
68	52	2022-04-19 15:01:33	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
69	53	2022-04-19 15:01:33	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
70	54	2022-04-19 15:01:33	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
71	55	2022-04-19 15:02:05	41,563.70 USD - Kraken - History - Ledger
72	56	2022-04-19 15:02:32	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
73	57	2022-04-19 15:02:44	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
74	37	2022-04-19 15:02:56	cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail

Appendix A4 — Relevant navigation/transition records

```
(kali@kali)~[/webhistory]
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "
SELECT
  v.id AS visit_id,
  (v.transition & 255) AS base_transition,
  v.from_visit,
  u.url
FROM visits v
JOIN urls u ON v.url = u.id
ORDER BY v.visit_time;
+-----+-----+-----+-----+
visit_id base_transition from_visit url
+-----+-----+-----+-----+
1         5         0 https://www.google.com/search?q=craigslist&rlz=1C10NGR_enUS989US9896oq=craigslist&saqs=chrome..69157j461131i1991433i465i512j0i402l2j0i512l3j0i1131i433l2j0i512l.2595j0j15sourced=chrome&ie=UTF-8
2         0         0 https://www.google.com/search?q=craigslist&rlz=1C10NGR_enUS989US9896oq=craigslist&saqs=chrome..69157j461131i1991433i465i512j0i402l2j0i512l3j0i1131i433l2j0i512l.2595j0j15sourced=chrome&ie=UTF-8
3         0         2 https://www.craigslist.org/

Session Actions Edit View Help
4         0         3 https://geo.craigslist.org/
5         0         4 https://baltimore.craigslist.org/
6         0         5 https://accounts.craigslist.org/login/home
7         0         6 https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L
8         7         7 https://accounts.craigslist.org/login/home
9         0         8 https://www.craigslist.org/
10        0         9 https://baltimore.craigslist.org/
11        0        10 https://post.craigslist.org/c/bal
12        0        11 https://post.craigslist.org/k/sHbOU0i_7BG68qUDIbeTbQ/bQTi5
13        0        12 https://post.craigslist.org/k/sHbOU0i_7BG68qUDIbeTbQ/bQTi5?s=type
14        7        13 https://post.craigslist.org/k/sHbOU0i_7BG68qUDIbeTbQ/bQTi5?s=cat
15        7        14 https://post.craigslist.org/k/sHbOU0i_7BG68qUDIbeTbQ/bQTi5?s=edit
16        7        15 https://post.craigslist.org/k/sHbOU0i_7BG68qUDIbeTbQ/bQTi5?s=geoverify
17        7        16 https://post.craigslist.org/k/sHbOU0i_7BG68qUDIbeTbQ/bQTi5?s=editimage
18        7        17 https://post.craigslist.org/k/sHbOU0i_7BG68qUDIbeTbQ/bQTi5?s=preview
19        7        18 https://post.craigslist.org/k/sHbOU0i_7BG68qUDIbeTbQ/bQTi5
20        0        19 https://post.craigslist.org/manage/7473121658
21        0         0 https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html
22        5         0 https://www.google.com/search?q=gmail&rlz=1C10NGR_enUS989US9896oq=gmail&saqs=chrome..69157j0i433i512l6j0i1131i433i512j0i433i512.1553j0j46sourced=chrome&ie=UTF-8
23        0         0 https://www.google.com/search?q=gmail&rlz=1C10NGR_enUS989US9896oq=gmail&saqs=chrome..69157j0i433i512l6j0i1131i433i512j0i433i512.1553j0j46sourced=chrome&ie=UTF-8

Session Actions Edit View Help
58        0        54 https://www.blockchain.com/btc/address/38RcsURWCmCbVocms2CnGz1FtD7x477mt
59        5         0 https://www.google.com/search?q=kraken&rlz=1C10NGR_enUS989US9896oq=kraken&saqs=chrome..69157j461671433j0i67j0i67j0i67j0i67j0i67j0i1131i433i512j0i512l2.2504j0j76sourced=chrome&ie=UTF-8
60        0         0 https://www.google.com/search?q=kraken&rlz=1C10NGR_enUS989US9896oq=kraken&saqs=chrome..69157j461671433j0i67j0i67j0i67j0i67j0i67j0i1131i433i512j0i512l2.2504j0j76sourced=chrome&ie=UTF-8
61        0        60 https://www.kraken.com/en-us/sign-in
62        0         0 https://www.kraken.com/en-us/sign-in
63        0         0 https://www.kraken.com/en-us/device-approval
64        0        63 https://www.kraken.com/u/trade?i=true
65        0        63 https://www.kraken.com/u/instant
66        0        63 https://www.kraken.com/u/funding
67        0         0 https://www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx%2F517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5
68        0        67 http://www.kraken.com/redirect?t=-_F22IhwssFD
69        0        68 https://www.kraken.com/redirect?t=-_F22IhwssFD
70        0        69 https://mempool.space/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5
71        0        63 https://www.kraken.com/u/history/ledger
72        0        48 https://mail.google.com/mail/u/0/#inbox/whctKXXXXrFqvtRjhGdkMBqVNFwtCnJjRvqLXsZnkFbrkGDZwdVPkdDvjBPxRxDLqKDSv?compose=JHrtfddgkfJdsFdvFrNvsxdZrPvcWRNCNnxcfgKzhGgsQLbhJdvrmBxZmtmQDrNtCmsDQGgsbKqsWGvLfLHKLJRFkQxXzGKQwpdVVBLCkGmPjnk
73        0        72 https://mail.google.com/mail/u/0/#inbox/whctKXXXXrFqvtRjhGdkMBqVNFwtCnJjRvqLXsZnkFbrkGDZwdVPkdDvjBPxRxDLqKDSv?compose=xQTrpQHKFJPCwtCvdCmpkQMLJJjqlLXwrvmVrRlfJzMF8KjnChwKnnVqVnLwLlQrtxpnZxkFwdTSMpnQJfQhFggKjB3xlGjFGbxgFTTBSfTzxLhpGtTp
74        0        73 https://mail.google.com/mail/u/0/#inbox/whctKXXXXrFqvtRjhGdkMBqVNFwtCnJjRvqLXsZnkFbrkGDZwdVPkdDvjBPxRxDLqKDSv

(kali@kali)~[/webhistory]
```

Appendix A5 — Suspected posting/advertisement evidence

```

kali@kali:~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite
SELECT
  v.id AS visit_id,
  datetime(v.visit_time/1000000 + strftime('%s', '1601-01-01'), 'unixepoch') AS visit_utc,
  u.title
FROM visits v
JOIN urls u ON v.url = u.id
WHERE u.title LIKE '%supplements%' OR u.url LIKE '%mail.google.com%';

```

visit_id	visit_utc	title
21	2022-04-19 13:58:31	cheaper than Rx supplements - health and beauty - by owner - ...
25	2022-04-19 13:58:40	Inbox (12) - unsub.fscs@gmail.com - Gmail
26	2022-04-19 13:58:40	Inbox (12) - unsub.fscs@gmail.com - Gmail
27	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email Google Workspace
28	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email Google Workspace
31	2022-04-19 13:58:43	Gmail
32	2022-04-19 13:58:43	Gmail
33	2022-04-19 13:58:43	Gmail
34	2022-04-19 13:58:44	Gmail
35	2022-04-19 13:58:44	Gmail
36	2022-04-19 13:59:26	Gmail
37	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
38	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
39	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
40	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
41	2022-04-19 13:59:29	Inbox (12) - unsub.fscs@gmail.com - Gmail
42	2022-04-19 13:59:31	Inbox (12) - unsub.fscs@gmail.com - Gmail
43	2022-04-19 14:04:12	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
44	2022-04-19 14:04:35	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
45	2022-04-19 14:04:46	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
46	2022-04-19 14:04:52	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
47	2022-04-19 14:05:59	Inbox (12) - unsub.fscs@gmail.com - Gmail
48	2022-04-19 14:56:47	cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail
72	2022-04-19 15:02:32	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
73	2022-04-19 15:02:44	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
74	2022-04-19 15:02:56	cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail

```

kali@kali:~/webhistory
$

```

Appendix A6 — Communication/payment evidence

```

kali@kali:~/webhistory
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite
SELECT
  v.id AS visit_id,
  datetime(v.visit_time/1000000 + strftime('%s', '1601-01-01'), 'unixepoch') AS visit_utc,
  u.url,
  u.title
FROM visits v
JOIN urls u ON v.url = u.id
WHERE u.url LIKE '%kraken%' OR u.url LIKE '%mempool%' OR u.url LIKE '%blockchain%';

```

visit_id	visit_utc	url	title
51	2022-04-19 14:58:59	https://www.google.com/search?q=blockchainexplorer&rlz=1C10NGR_enUS989US989eq=blockchainexplorer&baqs=chrome..69157j35139j01512L8.3445j0j76sourceid=chrome&ie=UTF-8	blockchain explorer - Google Search
52	2022-04-19 14:59:00	https://www.google.com/search?q=blockchainexplorer&rlz=1C10NGR_enUS989US989eq=blockchainexplorer&baqs=chrome..69157j35139j01512L8.3445j0j76sourceid=chrome&ie=UTF-8	blockchain explorer - Google Search
53	2022-04-19 14:59:01	https://www.blockchain.com/explorer	Blockchain Explorer - Search the Blockchain BTC ETH BCH
54	2022-04-19 14:59:02	https://www.blockchain.com/explorer	Blockchain Explorer - Search the Blockchain BTC ETH BCH
55	2022-04-19 14:59:06	https://www.blockchain.com/search?search=517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Blockchain Explorer
56	2022-04-19 14:59:06	https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Blockchain Explorer
57	2022-04-19 14:59:06	https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Blockchain Explorer
58	2022-04-19 14:59:21	https://www.blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1Ftd7x477mt	Address: 38RcsURWCDmCbYocmsZCnGz1Ftd7x477mt Blockchain Explorer
To direct input to this VM, move the mouse pointer inside or press Ctrl-G.			
59	2022-04-19 14:59:38	https://www.google.com/search?q=kraken&rlz=1C10NGR_enUS989US989eq=kraken&baqs=chrome..69157j461671433j0167j01671433j0167j01671433j0167j0113114331512j01512L2.2504j0j76sourceid=chrome&ie=UTF-8	kraken - Google Search
60	2022-04-19 14:59:39	https://www.google.com/search?q=kraken&rlz=1C10NGR_enUS989US989eq=kraken&baqs=chrome..69157j461671433j0167j01671433j0167j01671433j0167j0113114331512j01512L2.2504j0j76sourceid=chrome&ie=UTF-8	kraken - Google Search
61	2022-04-19 14:59:47	https://www.kraken.com/en-us/sign-in	Sign in to Kraken - Kraken Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)
62	2022-04-19 14:59:47	https://www.kraken.com/en-us/sign-in	Sign in to Kraken - Kraken Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)
63	2022-04-19 15:00:21	https://www.kraken.com/en-us/device-approval	Approve new device - Kraken Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)
64	2022-04-19 15:01:11	https://www.kraken.com/u/trade?i=true	
65	2022-04-19 15:01:12	https://www.kraken.com/u/instant	Kraken Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)
66	2022-04-19 15:01:21	https://www.kraken.com/u/funding	41,543.80 USD - Kraken - Funding - Overview
67	2022-04-19 15:01:31	https://www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx%2F517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
68	2022-04-19 15:01:33	http://www.kraken.com/redirect?t=-_F22IhwssFD	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
69	2022-04-19 15:01:33	https://www.kraken.com/redirect?t=-_F22IhwssFD	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
70	2022-04-19 15:01:33	https://mempool.space/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
71	2022-04-19 15:02:05	https://www.kraken.com/u/history/ledger	41,563.70 USD - Kraken - History - Ledger

```

kali@kali:~/webhistory
$

```

Appendix A7 — Download evidence and URL chain

```
(kali@kali)-[~/webhistory]
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "
SELECT
  d.id AS download_id,
  datetime(d.start_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS start_utc,
  d.target_path,
  d.received_bytes,
  c.url AS chain_url
FROM downloads d
LEFT JOIN downloads_url_chains c ON d.id = c.id;
"
```

download_id	start_utc	target_path	received_bytes	chain_url
1	2022-04-19 14:56:58	C:\Users\FSCS_User\Desktop\proof_of_payment.png	33844	https://i.imgur.com/dTgrkP7.png

```
(kali@kali)-[~/webhistory]
$
```

Appendix A8 — Final timeline and attribution correlation

```
(kali@kali)-[~/webhistory]
$ sqlite3 -header -column ~/Forensic_Evidence/Working_Copies/History_working_copy.sqlite "
SELECT
  v.id AS visit_id,
  datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,
  (v.transition & 255) AS trans,
  u.title
FROM visits v
JOIN urls u ON v.url = u.id
ORDER BY v.visit_time LIMIT 15;
"
```

visit_id	visit_utc	trans	title
1	2022-04-19 13:54:18	5	craigslist - Google Search
2	2022-04-19 13:54:19	0	craigslist - Google Search
3	2022-04-19 13:54:19	0	craigslist: baltimore jobs, apartments, for sale, services, community, and events
4	2022-04-19 13:54:19	0	craigslist: baltimore jobs, apartments, for sale, services, community, and events
5	2022-04-19 13:54:19	0	craigslist: baltimore jobs, apartments, for sale, services, community, and events
6	2022-04-19 13:54:22	0	craigslist account
7	2022-04-19 13:54:22	0	craigslist - account log in
8	2022-04-19 13:54:39	7	craigslist account
9	2022-04-19 13:55:03	0	craigslist: baltimore jobs, apartments, for sale, services, community, and events
10	2022-04-19 13:55:03	0	craigslist: baltimore jobs, apartments, for sale, services, community, and events
11	2022-04-19 13:55:05	0	baltimore choose type
12	2022-04-19 13:55:05	0	baltimore posting confirmation
13	2022-04-19 13:55:05	0	baltimore choose type
14	2022-04-19 13:55:10	7	baltimore choose category
15	2022-04-19 13:55:15	7	baltimore create posting

```
(kali@kali)-[~/webhistory]
$
```